

DELHI TECHNOLOGICAL UNIVERSITY

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ENVIRONMENTAL MANAGEMENT AND SUSTAINABLE DEVELOPMENT

Energy Conservation and Sustainability Proposal for Lighting Systems in Academic Blocks AB-3 & AB-4, Delhi Technological University

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Abstract

Our energy audit of the lighting in Academic Blocks AB-3 and AB-4 at Delhi Technological University (DTU) found a major opportunity. The core problem is that lights run 24/7, staying fully lit during the 14 non-operational hours from 6:00 PM to 8:00 AM.

This practice has a significant annual impact. The **7.728 kW** system load consumes **67,702 kWh** yearly, costing **₹609,319** and emitting **49.22 tonnes of CO₂ equivalent (tCO₂e)**.

A large portion of this is pure waste. The after-hours operation wastes **39,493 kWh**, costs **₹355,436**, and generates **28.71 tCO₂e** annually. This constant use also slashes the LED fixtures' lifespan from a potential **13.7 years** down to just **5.7 years**.

We propose installing a hybrid, sensor-based system. It would function normally during operational hours (**8:00 AM to 6:00 PM**) but switch to motion-activated control at night (**6:00 PM to 8:00 AM**). This ensures lights are only on when someone is present after hours. The total implementation cost is estimated at **₹264,000**.

The payoff is substantial. This solution would cut after-hours energy use by up to 95%, saving approximately **₹337,664** each year. That translates to a simple payback period of just **9.4 months**. It would also cut **27.27 tCO₂e** annually and extend the lights' calendar lifespan by 140%, significantly reducing maintenance and electronic waste.

We strongly recommend implementing this sensor solution. (A cheaper manual alternative was analyzed but is far less effective.) This initiative is a financially prudent and environmentally responsible investment for **Delhi Technological University**, aligning perfectly with its role as a leader in sustainable development.

1. Introduction: The Imperative for Energy Management in Academic Institutions

1.1 The Role of Universities in Driving Sustainability

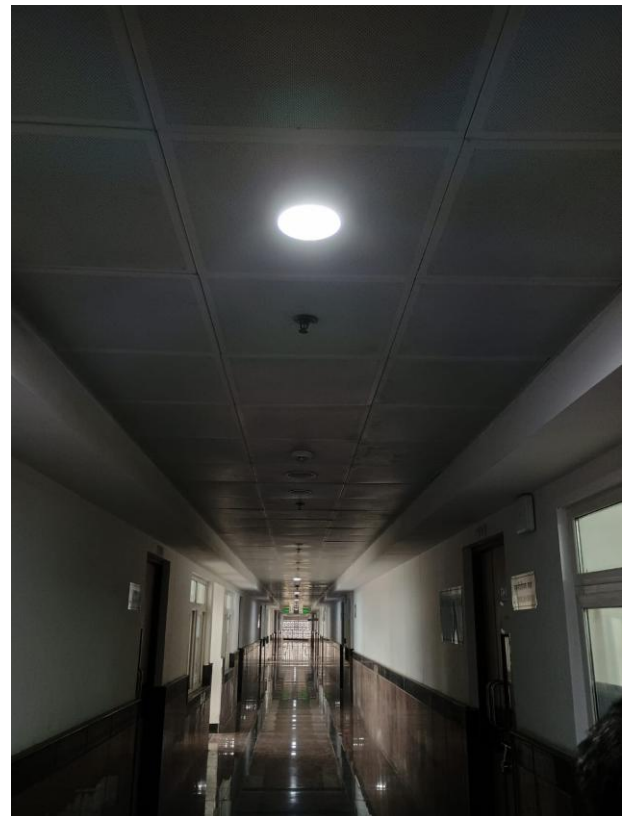
Educational institutions, particularly those with a focus on science and engineering like Delhi Technological University (DTU), serve a dual role in society. They are centers for knowledge dissemination and innovation, but they are also significant consumers of resources, operating as small cities with substantial energy footprints. In the contemporary context of climate change and resource scarcity, these institutions have a unique responsibility and opportunity to transition from being passive consumers to active demonstrators of sustainable practices. By transforming their campuses into "living laboratories," universities can provide tangible, real-world examples of environmental engineering principles in action.

This approach not only reduces operational costs and environmental impact but also enriches the educational experience for students. Case studies from various Indian educational institutions confirm a growing trend of implementing energy-efficient retrofits, such as LED lighting and automated controls, which have demonstrated significant reductions in energy consumption and carbon emissions, reinforcing the viability and importance of such initiatives.¹

1.2 Problem Statement: A Case of Unmanaged Energy Consumption at DTU

This investigation focuses on a specific and conspicuous instance of unmanaged energy consumption within the DTU campus. The two structurally identical Academic Blocks, **AB-3 and AB-4**, along with their connecting bridges, are subject to a lighting protocol that keeps corridor, balcony, and bridge lights operational 24 hours a day, **7 days a week**.

However, the official operational hours for these academic buildings are from **8:00 AM to 6:00 PM**. This creates a 14-hour window each day during which the buildings are closed for general academic purposes, yet the lighting systems remain fully energized. While potentially motivated by aesthetic considerations, this practice represents a continuous and unnecessary expenditure of electrical energy, contributing to a larger financial and environmental burden for the university.



1.3 Report Objectives and Scope

The primary objective of this report is to conduct a thorough engineering analysis of the aforementioned lighting systems to quantify the scale of this energy waste and propose a viable, data-driven solution. The specific goals are as follows:

1. To establish a quantitative baseline of the current annual energy consumption (kWh), associated financial costs (₹), and resultant carbon footprint (tCO_2e) of the specified lighting systems.
2. To isolate and precisely quantify the portion of this baseline that constitutes preventable waste, specifically the consumption occurring during the 14-hour non-operational period.
3. To propose technically feasible and economically sound solutions involving automated and procedural lighting controls to eliminate this waste.
4. To conduct a comprehensive impact analysis of the proposed solutions, evaluating their environmental benefits, financial returns, and operational advantages.

The scope of this analysis is strictly limited to the corridor and balcony lighting in AB-3 and AB-4 (from the basement to the 8th floor) and the lighting on the two connecting bridges. As per the project directive, lighting in safety-critical areas such as staircases and lift lobbies is explicitly excluded from this assessment to ensure that any proposed changes do not compromise campus safety standards.

2. Baseline Analysis: Quantifying the Current Energy and Environmental Footprint

To understand the magnitude of the energy conservation opportunity, it is first necessary to establish a comprehensive baseline of the current situation. This section details the physical assets involved and calculates their total annual energy consumption, financial cost, and carbon footprint under the existing 24/7 operational model.

2.1 Lighting Asset Inventory

A physical audit of the lighting fixtures in the specified areas of Academic Blocks AB-3, AB-4, and the connecting bridges was conducted. The inventory, detailing the type, quantity, and power rating of each fixture, is systematically presented in Table 1. This inventory forms the foundational dataset for all subsequent calculations in this report.



Table 1: Lighting Asset Inventory for AB-3, AB-4, and Connecting Bridges

Location	Fixture Type	Wattage per Fixture (W)	Quantity	Total Power (kW)
Academic Block 3 (AB-3)				
Basement Corridor	LED	12	26	0.312
Ground Floor Corridor	LED	12	26	0.312
Floors 1-8 Corridors	LED	12	240	2.880
<i>Subtotal for AB-3</i>			292	3.504
Academic Block 4 (AB-4)				
Basement Corridor	LED	12	26	0.312
Ground Floor Corridor	LED	12	26	0.312
Floors 1-8 Corridors	LED	12	240	2.880
<i>Subtotal for AB-4</i>			292	3.504
Connecting Bridges				
Bridge at 4th Floor	Tubelight	20	18	0.360

Bridge at 7th Floor	Tubelight	20	18	0.360
<i>Subtotal for Bridges</i>			36	0.720
Grand Total			620	7.728

The total connected lighting load for the areas under review is **7.728 kW**.

2.2 Current Annual Energy Consumption

With the total power load established, the annual energy consumption is calculated based on continuous, uninterrupted operation.

The formula for energy consumption is:

$$E = P \times t$$

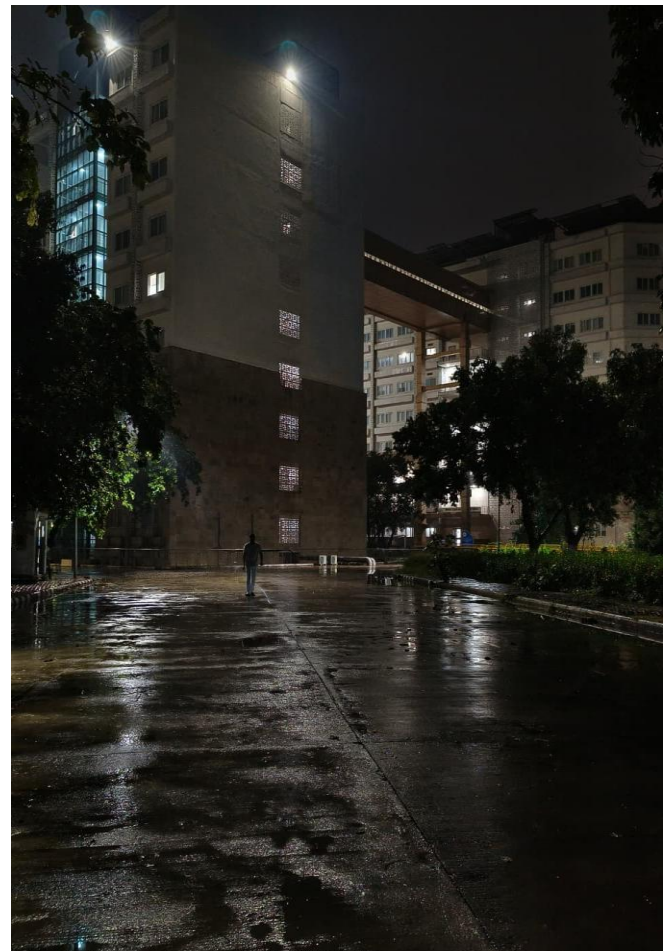
where:

- E is the energy consumed in kilowatt hours (kWh).
- P is the total power in kilowatts (kW).
- t is the time in hours (h).

Calculation:

- Total Power (P): 7.728 kW
- Hours of Operation per year (t):
24 hours/day $\times$ 365 days/year = 8760 hours/year

$$E_{\text{annual}} = 7.728 \text{ kW} \times 8760 \text{ h/year} = \mathbf{67,702.08 \text{ kWh/year}}$$



The total annual electricity consumption for the specified lighting is approximately **67,702 kWh**.

2.3 Current Annual Carbon Footprint

The environmental impact of this energy consumption is quantified by calculating the associated greenhouse gas emissions. This is achieved by multiplying the total energy consumption by the grid emission factor for India. The most recent and authoritative value is provided by the Central Electricity Authority (CEA) in its FY 2023-24 report, which states a weighted average emission factor of 0.727 tonnes of CO₂ per megawatt-hour (tCO₂/MWh), or 0.727 kilograms of CO₂ per kilowatt-hour (kg CO₂/kWh).⁴

The formula for carbon emissions is:

$$C = E_{\text{annual}} \times EF_{\text{grid}}$$

where:

- C is the total carbon dioxide emissions in kilograms (kg CO₂).
- E_{annual} is the annual energy consumption (kWh).
- EF_{grid} is the grid emission factor (kg CO₂/kWh).

Calculation:

$$C = 67,702.08 \text{ kWh/year} \times 0.727 \text{ kg CO}_2 / \text{kWh} = 49,219.41 \text{ kg CO}_2 / \text{year}$$

This is equivalent to **49.22 tonnes of CO₂** emitted annually.

2.4 Current Annual Financial Expenditure

The financial cost is determined using the commercial electricity tariff applicable in Delhi. As a large institutional consumer, DTU's energy consumption falls into the highest tariff slab. Therefore, any marginal change in consumption (i.e., savings) will be realized at this highest rate. Based on current tariff schedules for commercial entities in Delhi, a conservative rate of **₹9.00 per kWh** is used for this analysis.

The formula for annual cost is:

$$\text{Cost}_{\text{annual}} = E_{\text{annual}} \times \text{Tariff}$$

where:

- $\text{Cost}_{\text{annual}}$ is the total annual cost in Indian Rupees (₹).
- E_{annual} is the annual energy consumption (kWh).
- Tariff is the cost per unit of electricity (₹/kWh).

Calculation:

$$\text{Cost}_{\text{annual}} = 67,702.08 \text{ kWh/year} \times ₹9.00/\text{kWh} = ₹609,318.72 / \text{year}$$

The total annual expenditure on this unnecessary lighting is approximately ₹609,319.

3. The Anatomy of Waste: Isolating Non-Operational Consumption

The baseline analysis reveals the total impact of the current lighting protocol. This section dissects that data to isolate the specific component that constitutes preventable waste: the energy consumed during the 14-hour daily period when the academic blocks are non-operational. This analysis also extends beyond energy to consider the hidden operational costs associated with the accelerated degradation of physical assets.

3.1 Quantifying Wasted Resources

The period of waste is defined as the 14 **hours** between **6:00 PM and 8:00 AM** the following morning. The energy, cost, and emissions for this period are calculated as follows:

- Total Power (P): 7.728 kW
- Wasted Hours per year (t_{waste}): 14 hours/day $\times$ 365 days/year = **5110 hours/year**

Wasted Energy:

$$Energy_{waste} = 7.728kW \times 5110 \text{ hours/years} = \mathbf{39,492 \text{ kWh/year}}$$

Wasted Financial Expenditure:

$$Cost_{waste} = 39,492.88 \text{ kWh/year} \times 0.727 \text{ kg } CO_2/kWh = \mathbf{₹355,435.92/year}$$

Wasted Carbon Emissions:

$$C_{waste} = 39,492.88 \text{ kWh/year} \times 0.727 \text{ kg } CO_2/kWh = \mathbf{28,711.92 \text{ kg } CO_2/year}$$

In summary, of the total annual consumption, **39,493 kWh** of energy, **₹355,436** in cost, and **28.71 tonnes of CO₂** are attributable directly to operating the lights when the buildings are closed. This represents approximately 58% of the total lighting expenditure for these areas.

3.2 Impact on Asset Lifespan and Maintenance

A critical but often overlooked consequence of continuous operation is the accelerated depreciation of physical assets. Lighting fixtures have a finite operational lifespan, measured in hours. Unnecessary operation consumes this lifespan, leading to more frequent

replacements, increased maintenance labor, and higher volumes of electronic waste.

The average rated life for commercial LED fixtures is approximately 50,000 hours, while fluorescent tubelights are rated for a lifespan of around 24,000 to 36,000 hours; a conservative value of 30,000 hours is used for this analysis.⁸

The calendar lifespan is calculated as:

$$Life_{Years} = \frac{Rated\ Life(Hours)}{Annual\ Operating\ Hours}$$

Under the current 24/7 operation (8760 hours/year):

- **LED Lifespan:** $\frac{50,000\ hours}{8760\ hours/years} = 5.71\ years$
- **Tubelight Lifespan:** $\frac{30,000\ hours}{8760\ hours/years} = 3.42\ years$

- This calculation reveals that the university must budget for the complete replacement of all corridor tubelights every 3.5 years and all corridor LEDs every 5.7 years.
- This rapid replacement cycle incurs significant costs for both materials and labor and contributes to the university's electronic waste stream.
- More than half of the asset's useful life is being consumed during periods of zero utility, representing a significant hidden operational inefficiency.

4. A Strategic Solution: Automated and Procedural Lighting Control

Addressing the identified energy waste requires a shift from a manual, continuous-on protocol to an automated, schedule-based system. This section outlines two distinct solutions: a high-efficiency technology-based approach and a low-cost procedural alternative.

4.1 Primary Recommendation: Hybrid Motion Sensor System

The most effective and technologically advanced solution is the installation of a hybrid lighting control system that integrates timers with motion sensors. This system would operate in two distinct modes:

- **Operational Mode (8:00 AM - 6:00 PM):** The lights function as normal, remaining continuously on to ensure well-lit corridors for students and faculty.

- **Non-Operational Mode (6:00 PM - 8:00 AM):** The system switches to an occupancy-based mode. Lights automatically turn on only when motion is detected by a sensor and turn off after a preset period of inactivity.

Technology and Feasibility:

This system would utilize Passive Infrared (PIR) sensors, which are ideal for the long, narrow layout of corridors as they detect motion based on body heat and require a direct line of sight, minimizing false triggers from adjacent areas. These sensors are a mature and reliable technology, readily available from numerous suppliers. The system can be programmed to ensure hands-free convenience and enhanced safety for any personnel present after hours.

Implementation and Cost:

To ensure full coverage, it is estimated that two sensors would be required per floor for each academic block, and two sensors for each bridge.

- **Number of Sensors Required:** (10 floors/building $\times$ 2 buildings $\times$ 2 sensors/floor) + (2 bridges $\times$ 2 sensors/bridge) = **44 sensors**.
- **Estimated Cost:** Based on market rates for commercial-grade PIR sensors and professional installation, an estimated cost of **₹6,000 per installed unit** is used. This includes the sensor, control module, labor, and wiring.
- **Total Implementation Cost:** $44 \text{ sensors} \times ₹6,000/\text{sensor} = ₹264,000$

Projected Impact:

Occupancy in corridors during non-operational hours is extremely low. Therefore, a motion-activated system can achieve energy savings of up to 95% during this 14-hour window.

- **Annual Energy Savings:** $39,493 \text{ kWh/year} \times 95\% = 37,518 \text{ kWh/year}$
- **Annual Cost Savings:** $37,518 \text{ kWh/year} \times ₹9.00/\text{kWh} = ₹337,664/\text{year}$
- **Annual CO₂ Reduction:** $37,518 \text{ kWh/year} \times 0.727 \text{ kg CO}_2 = 27,276 \text{ kg CO}_2/\text{year}$ (or 27.28 tonnes)
- **Simple Payback Period:** $\frac{₹264,000}{₹337,664/\text{year}} \approx 0.78 \text{ years}$ (or 9.4 months).

4.2 Alternative Recommendation: Systematic Manual Alternation

A lower-technology, procedural alternative is to implement a systematic manual lighting schedule. This strategy involves illuminating only half of the lights during the non-operational period, alternating which lights are used on a daily basis.

- **On odd-numbered calendar days (1st, 3rd, 5th, etc.):** Only odd-numbered light fixtures in each corridor are turned on.
- **On even-numbered calendar days (2nd, 4th, 6th, etc.):** Only even-numbered light fixtures are turned on.

Implementation Analysis:

This approach has two potential implementation paths:

1. Procedural (No Cost):

This relies on maintenance or security staff to manually switch the correct circuits each evening. While it requires no capital investment, its effectiveness is entirely dependent on human diligence and is prone to error and inconsistency.

2. Partial Rewiring (High Cost):

For a reliable, automated solution, the lighting on each floor would need to be rewired into two separate "odd" and "even" circuits. These circuits could then be controlled by a simple alternating timer. This partial rewiring would be a complex and disruptive process, potentially involving significant labor and material costs.

Projected Impact:

Assuming that the system is implemented successfully (either manually or through rewiring), it would reduce energy consumption during the 14-hour non-operational period by exactly 50%.

- **Annual Energy Savings:** $39,493 \text{ kWh/year} \times 50\% = 19,746.5 \text{ kWh/year}$
- **Annual Cost Savings:** $19,746.5 \text{ kWh/year} \times ₹9.00/\text{kWh} = ₹177,718 \text{ /year}$
- **Annual CO₂ Reduction:** $19,746.5 \text{ kWh/year} \times 0.727 \text{ kg CO}_2/\text{kWh} = 14,356 \text{ kg CO}_2/\text{year}$ (or 14.36 tonnes).

While this method provides considerable savings over the current practice, it captures only half of the potential savings offered by the sensor-based solution and presents significant challenges in terms of reliability or implementation cost.

5. Projected Impact Analysis: A Triple-Bottom-Line Assessment

The implementation of the primary recommendation for a hybrid motion sensor system will yield significant and measurable benefits across three key domains: environmental protection, financial performance, and operational efficiency. This "triple-bottom-line" impact makes a compelling case for the project's approval.

5.1 Environmental Savings

The project's primary environmental benefit is the direct reduction of greenhouse gas emissions resulting from decreased electricity consumption from the grid. The annual energy savings are based on the 95% reduction calculated in Section 4.1.

- **Annual Energy Savings:** 37,518 kWh
- **Grid Emission Factor:** 0.727 kg CO₂/kWh

Projected Annual CO₂ Reduction:

$$C_{Reduction} = 37,518 \text{ kWh/year} \times 0.727 \text{ kg CO}_2/\text{kWh} = \mathbf{27,275.59 \text{ kg CO}_2/\text{year}}$$

The project will abate **27.28 tonnes of CO₂ equivalent emissions** each year, contributing directly to the reduction of DTU's overall carbon footprint.

5.2 Financial Savings and Investment Viability

The project represents a sound financial investment with a rapid return. The annual cost savings are equal to the value of the wasted energy, and the payback period is calculated by dividing the total investment by these annual savings. The complete cost-benefit analysis is presented in Table 2.

The analysis shows that the initial investment of **₹264,000** will be fully recovered through electricity bill savings in under ten months. After this point, the project will generate over **₹337,000** in net savings for the university every year for the lifespan of the equipment

Table 2: Cost-Benefit Analysis and Payback Period Calculation (Hybrid Sensor System)

Item	Description	Value
Investment Costs		
Motion Sensor Systems	44 units for ₹6,000 per installed unit	₹264,000
<i>Total Initial Investment</i>		₹264,000
Annual Returns		
Annual Energy Savings	Reduction of 37,518 kWh/year	
Annual Cost Savings	37,518 kWh/year × ₹9.00/kWh	₹337,664
Investment Viability		
Simple Payback Period (Years)	₹264,000 / ₹337,664	0.78 years
Simple Payback Period (Months)	0.78 years × 12 months/year	9.4 months

5.3 Operational Benefits

Beyond direct financial returns, the project delivers significant long-term operational advantages by preserving the lifespan of the lighting assets.

Extended Asset Lifespan:

By reducing the daily operational time from 24 hours to an effective average of approximately 10.7 hours (10 hours full operation + 5% of 14 hours non-operational), the annual operating hours decrease from 8760 to approximately 3905. This has a dramatic effect on the calendar lifespan of the fixtures.

Under the proposed hybrid sensor operation (~3905 hours/year):

- **New LED Lifespan:** $\frac{50,000 \text{ hours}}{3905 \text{ hours/year}} = 12.80 \text{ years}$
- **New Tubelight Lifespan:** $\frac{30,000 \text{ hours}}{3905 \text{ hours/year}} = 7.68 \text{ years}$

This represents a **124% increase** in the operational life for LEDs and a **125% increase** for tubelights. Instead of replacing LEDs every 5.7 years, they will now last for nearly 13 years. This extension directly translates into:

- **Reduced Maintenance:** The frequency of bulb replacements and associated labor costs will be more than halved.
- **Lower Procurement Costs:** The capital expenditure required for purchasing replacement fixtures will be deferred significantly, freeing up budget for other priorities.
- **Decreased E-Waste:** By extending the life of 620 fixtures, the project substantially reduces the volume of electronic waste the university generates over time, aligning with circular economy principles.

6. Conclusion and Formal Recommendations

This engineering analysis has systematically quantified a significant and easily rectifiable source of energy waste at the Delhi Technological University campus.

The continuous, 24/7 illumination of non-operational areas in Academic Blocks AB-3 and AB-4 is financially costly, environmentally burdensome, and operationally inefficient.

The data clearly indicates that an intelligent, automated solution can yield substantial and immediate returns.

The current lighting protocol results in an annual preventable waste of approximately **39,493 kWh**, costing **₹355,436** and emitting **28.71 tonnes of CO₂**.

The proposed primary solution, a hybrid motion sensor system, eliminates up to 95% of this waste with an investment that pays for itself in approximately 9.4 months.

The superiority of this solution over both the current practice and the manual alternation method is stark, as summarized in Table 3.

Table 3: Comparative Scenario Analysis (Annual Waste Reduction)

Metric	Scenario 1: Manual Alternation	Scenario 2: Hybrid Sensor System
Energy Savings (kWh)	19,747 (50%)	37,518 (95%)
Financial Savings (₹)	177,718	337,664
CO ₂ Reduction (tCO ₂)	14.36	27.28
Est. Upfront Cost (₹)	Zero (manual) to High (rewiring)	264,000
Reliability	Low (manual) to High (rewiring)	Very High

Furthermore, the proposed sensor solution delivers crucial operational benefits by extending the calendar lifespan of lighting assets by over 120%, thereby reducing long-term maintenance and procurement costs.

7. References

1. Patil, G. N., & Tanavade, S. S. (2024). Eco-friendly energy efficient classrooms and sustainable campus strategies: a case study on energy management and carbon footprint reduction. *International Journal of Energy Economics and Policy*, 14(3), 188-197.
2. *International Journal of Modern Engineering Research*. (2013). Energy Conservation in Lighting System by Replacing the Existing Illumination Scheme.
3. EQMag Pro. (2025). CEA Updates CO₂ Baseline Database.
4. Central Electricity Authority. (2024). CO₂ Baseline Database for the Indian Power Sector, User Guide Version 20.0.
5. NoBroker. (2025). Electricity Rate Per Unit in India.
6. NoBroker. (2025). Commercial Electricity Rate in Delhi.
7. Apex Lighting. (n.d.). Glossary: Average Rated Life.
8. Reddit. (2014). LED versus fluorescent lights.
9. Hykoont. (n.d.). How Long Do 4ft LED Bulbs Last?.
10. The Home Depot. (2025). Timer Switches.
11. Electrical Marketplace. (2025). Timers.

12. Bees Lighting. (2025). Light Switch Timers.
13. U.S. Department of Energy. (n.d.). Wireless Occupancy Sensors for Lighting Controls:
14. (PDF) ENERGY EFFICIENT LIGHTING DESIGN - A CASE STUDY OF JIGME NAMGYEL ENGINEERING COLLEGE - ResearchGate. (2025, November 3). Retrieved from [https://www.researchgate.net/publication/355673036_ENERGY_EFFICIENT_LIGHTING_DESIGN - A CASE STUDY OF JIGME NAMGYEL ENGINEERING COLLEGE](https://www.researchgate.net/publication/355673036_ENERGY_EFFICIENT_LIGHTING_DESIGN_-_A_CASE_STUDY_OF_JIGME_NAMGYEL_ENGINEERING_COLLEGE)
15. Eco-friendly energy efficient classrooms and sustainable campus strategies - ZBW. (2025, November 3). Retrieved from https://zbw.eu/econis-archiv/bitstream/11159/653635/1/1889840823_0.pdf
16. A Case Study on Energy Conservation in Campus Lighting in an Institution - International Journal of Modern Engineering Research. (2025, November 3). Retrieved from https://www.ijmer.com/papers/Vol3_Issue4/AN3419391941.pdf
17. Latest Commercial Electricity Rate in Delhi: Online and Offline Payment 2025 - NoBroker. (2025, November 3). Retrieved from <https://www.nobroker.in/blog/commercial-electricity-rate-in-delhi/>
18. Average Rated Life - Apex Lighting. (2025, November 3). Retrieved from <https://www.apexlighting.com/glossary/average-rated-life/>
19. LED versus Fluorescent Lights : r/engineering - Reddit. (2025, November 3). Retrieved from https://www.reddit.com/r/engineering/comments/2e17th/led_versus_fluorescent_lights/
20. How Long Do 4ft LED Bulbs Last? - Hykoont. (2025, November 3). Retrieved from <https://hykoont.com/blogs/news/how-long-do-4ft-led-bulbs-last>
21. Timer Switches - The Home Depot. (2025, November 3). Retrieved from <https://www.homedepot.com/b/Electrical-Wiring-Devices-Light-Controls-Timer-Switches/N-5yc1vZc334>
22. Discover the Convenience of Light Timers at Electrical Marketplace. (2025, November 3). Retrieved from <https://www.electricalmarketplace.com/collections/timers>
23. Timer Switches for Lights & Appliances - Bees Lighting. (2025, November 3). Retrieved from <https://www.beeslighting.com/collections/light-switch-timers>
24. Fact Sheet - Better Buildings Solution Center. (2025, November 3). Retrieved from <https://betterbuildingssolutioncenter.energy.gov/sites/default/files/attachments/Wireless-Sensors-Guidance.pdf>
25. PIR VS Microwave, How to Choose a Right Sensor - AGC Lighting. (2025, November 3). Retrieved from <https://www.agcled.com/blog/pir-vs-microwave-how-to-choose-a-right-sensor.html>
26. Microwave Motion Sensor vs. PIR Motion Sensor Lighting - LED Light Expert. (2025, November 3). Retrieved from <https://www.ledlightexpert.com/Microwave-Motion-Sensor-vs-PIR-Motion-Sensor-Lighting- b 136.html>
27. Long-Range Motion Sensors Explained: PIR vs Microwave, Timers & Load Ratings. (2025, November 3). Retrieved from <https://www.futurelight.co.za/blogs/news/long-range-motion-sensors-explained-pir-vs-microwave-timers-load-ratings>